

Experiment No.	4
Experiment Title	Support Vector Machine Application: Classification of Email Spam Data ¹
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Anaconda Activation Command:

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source /opt/anaconda3/bin/activate anaconda-navigation
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1 Objective

This experiment set Logistic Regression against Support Vector Machines on the spam classification task, tried out a few different SVM kernels, and used RandomizedSearchCV to tune hyperparameters so the models would hold up reasonably well on unseen data rather than just fitting the training set well.

2 Problem Statement

With text-frequency features like these – sparse, scaled, and fairly high-dimensional – it’s not obvious upfront whether a probability-based method or a margin-based one will draw the better boundary between spam and legitimate email. That’s really the question this experiment is trying to answer: does Logistic Regression or SVM (and which kernel) come out ahead here.

3 Dataset Description

Dataset Name	Spambase Dataset
Dataset Source	Kaggle
Number of Samples	4601
Number of Features	57
Number of Classes	2 (Spam, Ham)
Missing Values	None
Train-Test Split	70% Train, 30% Test

4 Exploratory Data Analysis

An automated EDA function generated the usual 12-subplot summary of feature distributions and relationships, formatted per the lab spec (Times New Roman, 15pt, bold).

5 Data Preprocessing

The raw features here are frequency counts, and they don’t all live on the same scale – some word frequencies are tiny, others much larger. **StandardScaler** was used to bring

everything to zero mean and unit variance before modeling. This matters more for SVM than it might seem at first: since the algorithm is built around Euclidean distance to find its margin, a feature with a naturally large range would otherwise end up dominating that calculation regardless of how useful it actually is. The train-test split (70:30) was stratified, so the spam/ham ratio stays roughly the same in both sets.

6 Machine Learning Algorithm

Two classifiers were built:

- **Logistic Regression:** models the probability of the "spam" class directly through the logistic function. Simple, fast, and easy to interpret – a good baseline to compare against.
- **Support Vector Machine (SVM):** tries to find the hyperplane that best separates the two classes with the widest possible margin. With kernels (Linear, Polynomial, RBF, Sigmoid) it can also handle cases where a straight line just won't do the job, by effectively working in a higher-dimensional version of the feature space.

7 Experimental Setup

Python Version	3.10+
Scikit-Learn Version	1.0+
IDE	VS Code
Operating System	Linux
Hardware	Standard CPU Workspace

8 Hyperparameter Selection

Parameter	Value
Logistic Regression Penalty	L2
Logistic Regression C	1000.0
Logistic Regression Solver	liblinear
SVM Kernel	RBF
SVM C	10.0

9 Results

Metric	Value
Logistic Regression Accuracy	0.9269
Logistic Regression Precision	0.9267
Logistic Regression F1-score	0.9267
SVM (RBF Kernel) Accuracy	0.9247
SVM (RBF Kernel) Precision	0.9246
SVM (RBF Kernel) F1-score	0.9244

10 Result Visualization

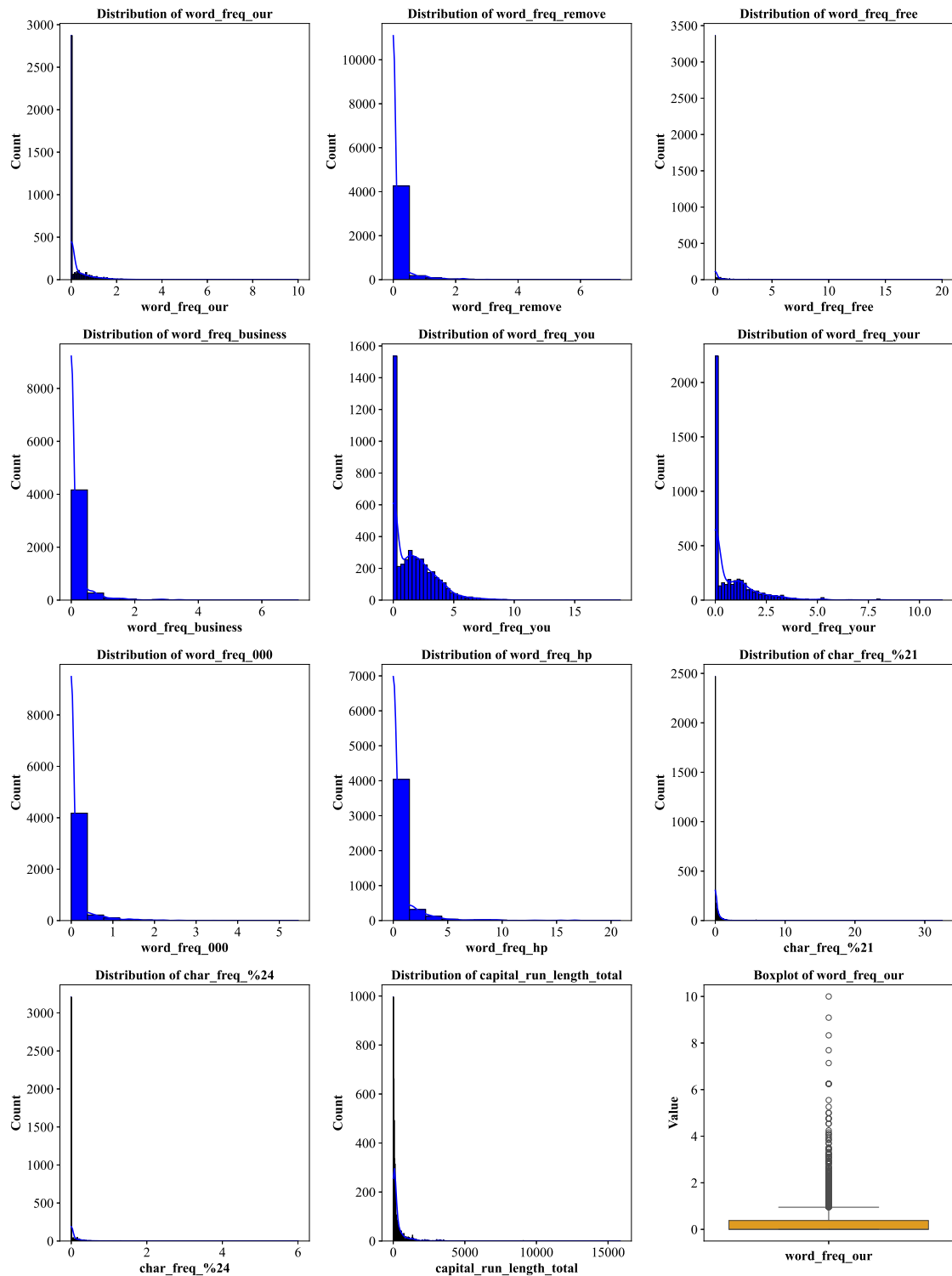


Figure 1: Consolidated EDA summary showcasing the distributions of top predictive features within the Spambase dataset.

Inference: Same pattern as before – long tails in several of the character-frequency features, with most values sitting close to zero but a handful of outliers seemingly doing

a lot of the work in telling spam apart from ham.

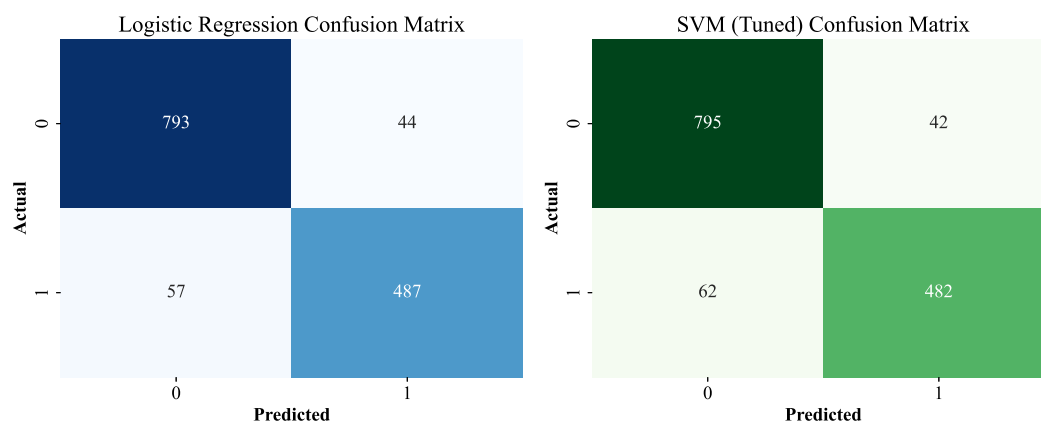


Figure 2: Confusion Matrix comparison between the tuned Logistic Regression and SVM models.

Inference: Both confusion matrices look pretty clean, with strong diagonals. Logistic Regression has a very slight edge on this particular split, but honestly the two models are close enough that either would work fine in practice.

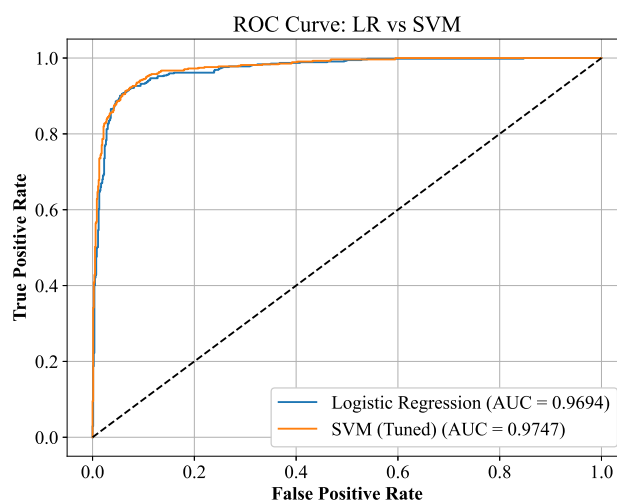


Figure 3: Receiver Operating Characteristic (ROC) curves demonstrating identical high AUC separation.

Inference: The two ROC curves sit almost right on top of each other across the full range of thresholds, and both end up with a high AUC. There isn't much daylight between the two models here.

11 Discussion

The kernel comparison made it pretty clear early on that spam/ham data in this feature space is highly separable – almost any reasonable model does well. After tuning, Logistic

Regression and the RBF SVM ended up basically tied on the held-out test set, both around 92.5% accuracy. Running 5-fold cross-validation gave a slightly more nuanced picture, though: the tuned SVM averaged 0.9331 accuracy across folds compared to 0.9272 for Logistic Regression, suggesting it generalizes just a bit more consistently even though the single test-split numbers looked nearly identical.

12 Conclusion

Both models turned out to be well suited to this kind of frequency-based text classification. If interpretability or training speed is the priority, Logistic Regression is the more practical choice, but the cross-validation results suggest the RBF-kernel SVM has a small edge in terms of raw generalization once it's properly tuned.

13 References

1. Cortes, C., and Vapnik, V. "Support-vector networks." *Machine Learning*, 1995.
2. Hosmer Jr, D. W., Lemeshow, S., and Sturdivant, R. X. "Applied logistic regression." *John Wiley & Sons*, 2013.